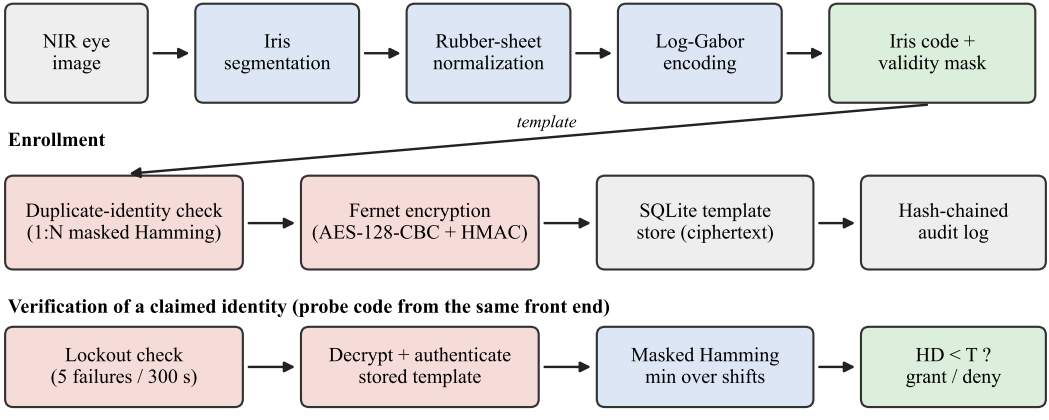
NIR eye image
Iris segmentation
Rubber-sheet normalization
Log-Gabor encoding
Iris code + validity mask
template
Enrollment
Duplicate-identity check (1:N masked Hamming)
Fernet encryption (AES-128-CBC + HMAC)
SQLite template store (ciphertext)
Hash-chained audit log
Verification of a claimed identity (probe code from the same front end)
Lockout check (5 failures / 300 s)
Decrypt + authenticate stored template
Masked Hamming min over shifts
HD < T ? grant / deny